

Higher strong-drive terms and the compensated-to-essential connection

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1 Displaced oscillator

For positive drive,

$$H/V = \frac{1}{2}a^{\dagger 2}a^2 + \delta a^{\dagger}a + \eta(a^{\dagger} + a), \quad a = b - r, \quad r^3 + \delta r = \eta.$$

Expanding the already normally ordered factors $(b^{\dagger} - r)^2(b - r)^2$, and then using stationarity, gives the exact identity

$$H/V = -\frac{3}{2}r^4 - \delta r^2 + (2r^2 + \delta)b^{\dagger}b + \frac{1}{2}r^2(b^{\dagger 2} + b^2) - r(b^{\dagger 2}b + b^{\dagger}b^2) + \frac{1}{2}b^{\dagger 2}b^2. \quad (1)$$

The real Bogoliubov transformation $b = Cc + Sc^{\dagger}$ has

$$\tanh(2\theta) = -\frac{r^2}{2r^2 + \delta}, \quad C = \cosh \theta, \quad S = \sinh \theta, \quad \Omega^2 = (2r^2 + \delta)^2 - r^4.$$

It diagonalizes the quadratic part and gives

$$E_n^{(2)}/V = -\frac{3}{2}r^4 - \delta r^2 + \Omega(n + \frac{1}{2}) - \frac{1}{2}(2r^2 + \delta).$$

2 Order-one interaction energy

At leading order $C^2 + S^2 = 2/\sqrt{3}$, $2CS = -1/\sqrt{3}$. The cubic vertex is odd in $c, c^{\dagger}$, so its first-order expectation vanishes. Exact ladder algebra gives

$$\Delta_{4,n}^{(1)} = \frac{18n^2 + (18 - 16\sqrt{3})n + 15 - 8\sqrt{3}}{24}, \quad (2)$$

$$\Delta_{3,n}^{(2)} = -\frac{30n^2 + (30 - 24\sqrt{3})n + 23 - 12\sqrt{3}}{36}. \quad (3)$$

The second expression is the finite sum over intermediate shifts $n' - n = \pm 1, \pm 3$, with denominator $\sqrt{3}(n - n')$. Their sum simplifies to

$$\Delta_{4,n}^{(1)} + \Delta_{3,n}^{(2)} = -\frac{6n^2 + 6n + 1}{72}.$$

Solving the stationarity equation sufficiently far gives

$$r = \eta^{1/3} - \frac{\delta}{3}\eta^{-1/3} + \frac{\delta^3}{81}\eta^{-5/3} + O(\eta^{-7/3}).$$

The exact quadratic energy contributes $\delta(1 - 2\delta)/6$ at order one. Therefore formal fixed-level Rayleigh–Schrödinger perturbation theory yields

$$\boxed{\frac{E_n}{V} = -\frac{3}{2}\eta^{4/3} + \left[\delta + \sqrt{3}(n + \frac{1}{2}) - 1\right]\eta^{2/3} + \frac{\delta(1 - 2\delta)}{6} - \frac{6n^2 + 6n + 1}{72} + O(\eta^{-2/3})}. \quad (4)$$

In particular,

$$\frac{E_{n+1} - E_n}{V} = \sqrt{3}\eta^{2/3} - \frac{n+1}{6} + O(\eta^{-2/3}), \quad \frac{E_{n+2} - 2E_{n+1} + E_n}{V} = -\frac{1}{6} + O(\eta^{-2/3}).$$

The next coefficient has not been evaluated. Its quadratic contribution is

$$\frac{\delta^3}{27} + \frac{\sqrt{3}\delta^2 n}{18} - \frac{\delta^2}{9} + \frac{\sqrt{3}\delta^2}{36}.$$

It must be supplemented by finite- r vertex and denominator corrections, second-order V_4^2 , third-order $V_3^2 V_4$ including the normalization subtraction, and fourth-order V_3^4 including its subtraction terms. Terms with an odd total number of cubic vertices vanish by parity. Thus the unfinished calculation is an explicit finite collection of ladder sums. Indeed, put $X_{jk} = \langle j|V_3/r|k\rangle$, $Y_{jk} = \langle j|V_4|k\rangle$, and $D_k = \sqrt{3}(n-k)$. Its unevaluated interaction part is exactly

$$\begin{aligned} & \sum_{k \neq n} \frac{Y_{nk} Y_{kn}}{D_k} + \sum_{k, l \neq n} \frac{X_{nk} X_{kl} Y_{ln} + X_{nk} Y_{kl} X_{ln} + Y_{nk} X_{kl} X_{ln}}{D_k D_l} \\ & - Y_{nn} \sum_{k \neq n} \frac{X_{nk} X_{kn}}{D_k^2} + \sum_{k, l, m \neq n} \frac{X_{nk} X_{kl} X_{lm} X_{mn}}{D_k D_l D_m} \\ & - \left(\sum_{k \neq n} \frac{X_{nk} X_{kn}}{D_k} \right) \left(\sum_{l \neq n} \frac{X_{nl} X_{ln}}{D_l^2} \right). \end{aligned}$$

The sums terminate because X shifts by $\pm 1, \pm 3$ and Y by $0, \pm 2, \pm 4$. Adding the finite- r derivatives of the lower-order terms and simplifying this expression is the smallest remaining algebraic task.

3 Complete pullback to the Bargmann function

Let $z = t^2$, $t = \eta^{1/6}x$, $u = \eta^{2/3}$, $B_1 = 2\eta$, and $B_2 = 2\delta$. The exact covering substitution and Liouville transformation combine to

$$\boxed{\mathcal{T}_u W = \Psi(z) = t^{1/2-B_2} \exp\left(\frac{B_1}{2t^2}\right) \left(\frac{d\zeta}{dx}\right)^{-1/2} W(\zeta(x)).} \quad (5)$$

The endpoint action and Jacobian are

$$u \int^x \sqrt{Q_u(s)} ds = -\frac{u}{x^2} - 2(1-\delta) \log x + O(1), \quad \zeta \sim \frac{\pm i\sqrt{2}}{x}, \quad \left(\frac{d\zeta}{dx}\right)^{-1/2} \sim cx.$$

Choose neutral labels in one consistent sector:

$$W_1 \sim x^{-3/2+2\delta} e^{-u/x^2}, \quad W_2 \sim x^{5/2-2\delta} e^{u/x^2}.$$

Because the gauge exponential in (5) is e^{u/x^2} , all powers and exponentials multiply explicitly to

$$\mathcal{T}_u W_1 \sim 1, \quad \mathcal{T}_u W_2 \sim x^{4-4\delta} e^{2u/x^2} \sim z^{2-2\delta} e^{2\eta/z}.$$

Changing the action branch exchanges the labels, not the cancellation versus enhancement distinction. A generic continuation therefore has

$$\Psi(z; E) \sim C_1(E)[1 + \dots] + C_{\text{ess}}(E) z^{2-2\delta} e^{2\eta/z} [1 + \dots].$$

The cancellation is not automatic, and continuation can reintroduce the second component.

4 Connection conjecture

Conjecture 1 (Global compensated-to-essential coefficient). *There is a finite physical canonical domain chain and normalized endpoint bases such that*

$$\mathcal{T}_u W_1^{(j)} \sim 1, \quad \mathcal{T}_u W_2^{(j)} \sim z^{2-2\delta} e^{2\eta/z},$$

and

$$\mathcal{A}_{\Gamma_{\text{phys}}} W_1^{(0)} = C_1 W_1^{(m)} + C_{\text{ess}} W_2^{(m)}.$$

With the displayed Wronskian order,

$$C_{\text{ess}} = \frac{\mathcal{W}_{\zeta}(\mathcal{A}_{\Gamma_{\text{phys}}} W_1^{(0)}, W_1^{(m)})}{\mathcal{W}_{\zeta}(W_2^{(m)}, W_1^{(m)})}.$$

Physical symmetries reduce holomorphic patching to $C_{\text{ess}} = 0$. Its zeros with multiplicities equal those of $\Delta_{\text{WI}}(2\eta, 2\delta, -2E/V, 2\eta)$; locally $C_{\text{ess}} = G\Delta_{\text{WI}}$, with G holomorphic and nonzero. Near a fixed level,

$$C_{\text{ess}} = \frac{A}{\Gamma[-p_u(E)]} + R, \quad A \neq 0, \quad R = O(u^{-1}A).$$

Every existence, equivalence, symmetry-reduction, and error assertion in this conjecture remains unproved. Nonzero basis rescalings multiply C_{ess} by a nonzero factor and preserve its zero set. Formula (4) has an independent formal operator derivation and does not prove the conjecture.

5 Status table

Status	Statement
Exact	Shifted identity (1), Bogoliubov frequency, pullback (5), and symbolic ladder matrix elements.
Formal	Fixed- n Rayleigh–Schrödinger expansion through $O(1)$ and the endpoint cancellation/enhancement asymptotics.
Conjectured	Canonical domain chain, $C_{\text{ess}} = 0$, determinant equivalence, and $O(u^{-1})$ connection remainder.
Open	Rigorous perturbative remainder, global progressive paths, and the complete $\eta^{-2/3}$ coefficient.